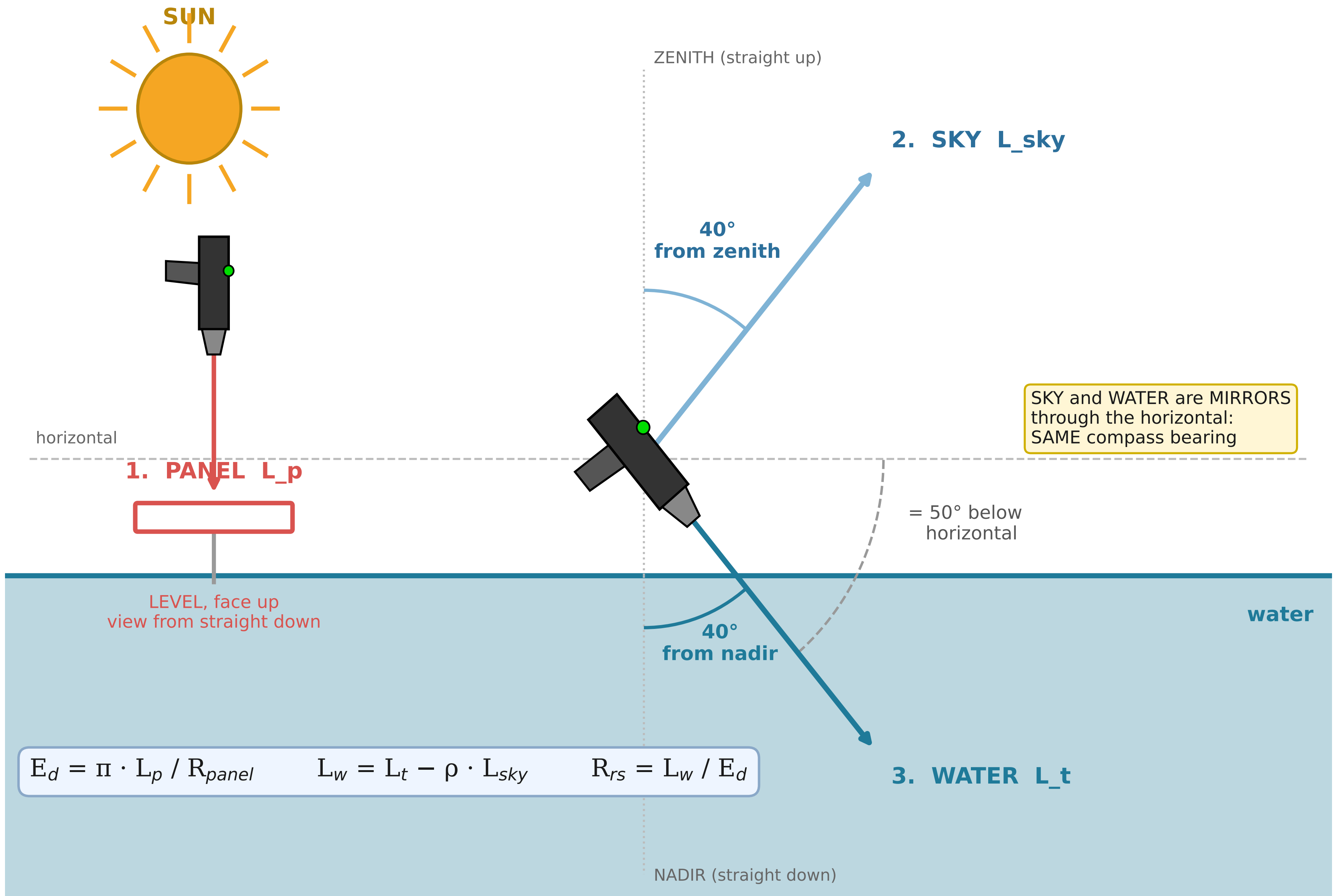
ABOVE-WATER R_{rs} — three-scan field method · Spectral Evolution NaturaSpec Plus

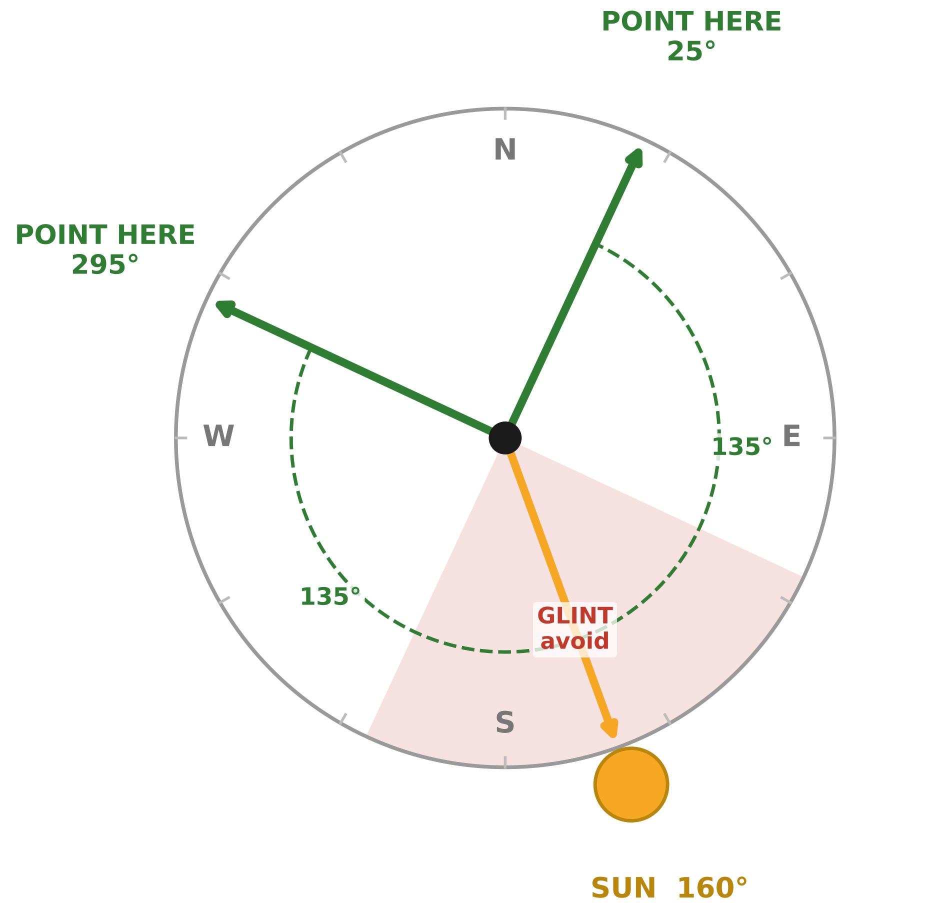
Geometry after Mobley (1999): view 40° from nadir, 135° in azimuth from the sun. Sky scan is the MIRROR of the water scan.

A. SIDE VIEW — the three scans and their angles



$$E_d = \pi \cdot L_p / R_{panel} \qquad L_w = L_t - \rho \cdot L_{sky} \qquad R_{rs} = L_w / E_d$$

B. PLAN VIEW — which way to point (looking down)



Both bearings are 135° in azimuth from the sun. Pick whichever is clear of the boat, its shadow and its wake. The GUI computes these for you from latitude, longitude, date and time.

1. PANEL

White reference panel (Spectralon), LEVEL and face up, in full sun. No shadow on it, yours included. View from straight down. Do not touch the white face.

DARWin: take as the REFERENCE. Re-take every ~10 min, and the moment the light changes.

2. SKY

Sky at 40° from ZENITH = 50° ABOVE horizontal.

SAME compass bearing as the water scan. Not the opposite one. This is the single most common field mistake.

DARWin: TARGET scan. Save.

3. WATER

Water at 40° from NADIR = 50° BELOW horizontal, 135° in azimuth from the sun.

No visible glint, no foam or whitecaps, no boat shadow, no wake.

DARWin: TARGET scan. Save. Repeat 2 and 3 five times.

RECORD

WIND SPEED (m/s) — sets ρ,
  the largest error in the whole
  measurement. ρ = 0.028 is only
  valid below ~5 m/s.
Cloud, or sun obscured?
Time + lat/lon → solar angle
Water clear or turbid?
  turbid → do NOT use nir_zero
Panel reflectance used (0.99?)

CHECK ON THE SPOT

Run the GUI at the station.

R_rs positive across 400-700?
  negative = over-subtracted
Peak where the water looks?
  blue = clear, green = turbid
Replicates lie on top of
  each other?
R_rs(443) order 1e-3 to 1e-2?

Copy the .sed files off tonight.